Project Design Phase-II

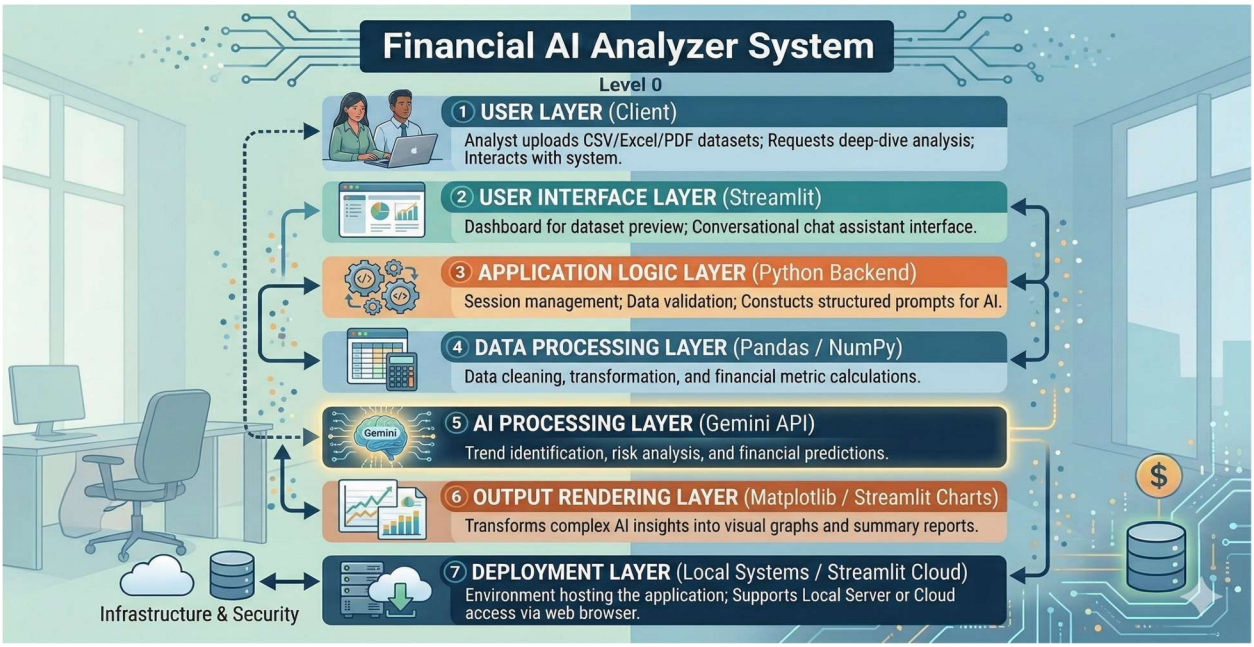
Technology Stack (Architecture & Stack)

Date	08 Feb 2026
Team ID	LTVIP2026TMIDS84828
Project Name	Financial AI Analyzer – Transforming Complex Data into Actionable Insights
Maximum Marks	4 Marks

Technical Architecture

The Financial AI Analyzer follows a Client–AI Architecture Model consisting of the following layers:

1. **User Interface Layer :**Web interface where users upload financial datasets and interact with the AI assistant.
2. **Application Logic Layer :** Python-based backend that processes financial data and manages system operations.
3. **AI Processing Layer :**AI model that analyzes financial data and generates insights.
4. **Output Rendering Layer :**Displays structured financial insights through charts, summaries, and dashboards.
5. **Deployment Layer :**System deployed locally or on a cloud platform for user access.



The deliverable includes the architecture diagram along with the information shown in Table-1 and Table-2.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based interface for uploading financial datasets and interacting with AI insights	Streamlit (Python-based Web Framework)
2	Application Logic-1	Financial dataset processing and data preparation	Python
3	Application Logic-2	AI-based financial analysis and prompt generation	Python
4	Application Logic-3	Session management and switching between analysis and assistant modes	Streamlit Session State
5	File Storage	Temporary storage of uploaded financial datasets during processing	Local Runtime Memory / Pandas DataFrames
6	External API-1	AI-powered financial analysis and natural language processing	Gemini API / Financial AI Model
7	Machine Learning Model	AI model for financial trend analysis, prediction, and insight generation	Generative AI Model
8	Infrastructure (Server / Cloud)	Local development and optional cloud deployment	Local System / Streamlit Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Application built using open-source libraries and frameworks	Python, Streamlit, Pandas, Matplotlib, python-dotenv

S.No	Characteristics	Description	Technology
2	Security Implementations	Secure API key handling using environment variables without exposing credentials	.env file, Streamlit Secrets, OS Environment Variables
3	Scalable Architecture	Modular architecture separating UI, backend logic, and AI analysis layers	API-Based AI Integration
4	Availability	Web-based deployment accessible through browsers depending on server availability	Streamlit Cloud / Local Server
5	Performance	Optimized financial data processing and AI analysis for faster response time	AI Model Processing with optimized prompts